

Homework 6

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Physics 16

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Question 1. Purcell chapter 6, exercise 6.38 (Uniform magnetic field in off-center hole.) A cylindrical rod with radius R carries current I (with uniform current density), with its axis lying along the z axis. A cylindrical cavity with an arbitrary radius is hollowed out from the rod at an arbitrary location; a cross section is shown in Purcell Figure 6.44. Assume that the current density in the remaining part stays the same (which would be the case for a fixed voltage difference between the ends). Let $\vec{a}$ be the position of the center of the cavity with respect to the center of the rod. Show that the magnetic field inside the cylindrical cavity is uniform (in both magnitude and direction). Hint: Show that the field inside a solid cylinder can be written in the form $\vec{B} = (\mu_0 I / 2\pi R^2) \hat{z} \times \vec{r}$, and then use superposition with another appropriately chosen cylinder.

Solution. First, we establish the field inside a solid cylinder carrying uniform current density $J = I/(\pi R^2)$. By Ampère's law with a circular loop of radius $r < R$,

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}.$$

J is uniform, so the enclosed current is proportional to the enclosed area

$$I_{\text{enc}} = J \cdot \pi r^2 = \frac{I}{\pi R^2} \cdot \pi r^2 = I \frac{r^2}{R^2}.$$

By symmetry, $\vec{B}$ is tangential and constant on the loop, so

$$|\vec{B}|(2\pi r) = \mu_0 I \frac{r^2}{R^2} \implies |\vec{B}| = \frac{\mu_0 I r}{2\pi R^2}.$$

Now, we establish the direction. The field points in the $\hat{\phi}$ direction (tangential), which we can express using a cross product. Note that $\hat{z} \times \hat{r} = \hat{\phi}$, so

$$\vec{B} = B \hat{\phi} = \frac{\mu_0 I r}{2\pi R^2} \hat{\phi} = \boxed{\frac{\mu_0 I}{2\pi R^2} \hat{z} \times \vec{r} = \frac{\mu_0 J}{2} \hat{z} \times \vec{r}}$$

Now consider the cylinder with the cavity. We treat the cavity to have current density $-J$, and because $\vec{r} - \vec{a}$ is the position measured from the center of the cavity, we apply the same equation to get

$$\vec{B}_{\text{cavity}} = \frac{\mu_0 J}{2} \hat{z} \times (\vec{r} - \vec{a}),$$

Therefore,

$$\vec{B}_{\text{total}} = \vec{B}_{\text{cylinder}} - \vec{B}_{\text{cavity}} = \frac{\mu_0 J}{2} \hat{z} \times \vec{r} - \frac{\mu_0 J}{2} \hat{z} \times (\vec{r} - \vec{a}) = \frac{\mu_0 J}{2} \hat{z} \times [\vec{r} - (\vec{r} - \vec{a})]$$

$$= \boxed{\vec{B}_{\text{total}} = \frac{\mu_0 J}{2} \hat{z} \times \vec{a} = \frac{\mu_0 I}{2\pi R^2} \hat{z} \times \vec{a},}$$

□

Question 2. Purcell chapter 6, exercise 6.39 (Constant magnitude of B .) How should the current density inside a thick cylindrical wire depend on r so that the magnetic field has constant magnitude inside the wire?

Solution. By Ampère's law with a circular loop of radius r :

$$B(2\pi r) = \mu_0 I_{\text{enc}}(r) = \mu_0 \int_0^r J(r') 2\pi r' dr'.$$

For B to be constant, the right side must be proportional to r :

$$\int_0^r J(r') 2\pi r' dr' = \alpha r,$$

for some constant α . Differentiating both sides with respect to r :

$$J(r) \cdot 2\pi r = \alpha \implies J(r) = \frac{\alpha}{2\pi r}.$$

Therefore,

$$\boxed{J(r) \propto \frac{1}{r}.$$

□

Question 3. A coaxial cable consists of a long cylindrical copper wire of radius r_1 surrounded by a cylindrical shell of inner radius r_2 and outer radius r_3 . The wire and the shell carry equal and opposite currents I uniformly distributed over their volumes. Find the magnetic field in each of the following regions: $r < r_1$, $r_1 < r < r_2$, $r_2 < r < r_3$, and $r > r_3$.

Solution. Apply Ampère's law, $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}$, with a circular Amperian loop of radius r in each region. When $r < r_1$, the current density in the wire is $J = I/(\pi r_1^2)$, so

$$I_{\text{enc}} = J \cdot \pi r^2 = I \frac{r^2}{r_1^2}.$$

Therefore,

$$B(2\pi r) = \mu_0 I \frac{r^2}{r_1^2} \implies \boxed{\vec{B} = \frac{\mu_0 I r}{2\pi r_1^2} \hat{\phi}.}$$

When $r_1 < r < r_2$, all of the inner wire's current is enclosed, and no current from the shell, so

$$I_{\text{enc}} = I.$$

Therefore,

$$B(2\pi r) = \mu_0 I \implies \boxed{\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi}.}$$

When $r_2 < r < r_3$, the shell carries current $-I$ (opposite direction) with uniform current density $J_s = I/[\pi(r_3^2 - r_2^2)]$. The enclosed current from the shell is

$$I_{\text{shell, enc}} = -J_s \cdot \pi(r^2 - r_2^2) = -I \frac{r^2 - r_2^2}{r_3^2 - r_2^2}.$$

Adding the full current $+I$ from the inner wire,

$$I_{\text{enc}} = I - I \frac{r^2 - r_2^2}{r_3^2 - r_2^2} = I \left(\frac{r_3^2 - r^2}{r_3^2 - r_2^2} \right).$$

Therefore,

$$B(2\pi r) = \mu_0 I \left(\frac{r_3^2 - r^2}{r_3^2 - r_2^2} \right) \implies \boxed{\vec{B} = \frac{\mu_0 I}{2\pi r} \cdot \frac{r_3^2 - r^2}{r_3^2 - r_2^2} \hat{\phi}.}$$

When $r > r_3$, the inner wire contributes $+I$ and the outer shell contributes $-I$, so

$$I_{\text{enc}} = I - I = 0.$$

Therefore,

$$\boxed{\vec{B} = 0.}$$

□

Question 4. Purcell chapter 6, exercise 6.43 (Vector potential inside a wire.) A round wire of radius r_0 carries a current I distributed uniformly over the cross section of the wire. Let the axis of the wire be the z axis, with $\hat{z}$ the direction of the current. Show that a vector potential of the form $\vec{A} = A_0 \hat{z}(x^2 + y^2)$ will correctly give the magnetic field $\vec{B}$ of this current at all points inside the wire. What is the value of the constant A_0 ?

Solution. We have $\vec{A} = A_0(x^2 + y^2) \hat{z}$, so $A_x = 0$, $A_y = 0$, and $A_z = A_0(x^2 + y^2)$. Hence,

$$\begin{aligned}\vec{B} &= \nabla \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ 0 & 0 & A_0(x^2 + y^2) \end{vmatrix} \\ &= \hat{x} \left(\frac{\partial A_z}{\partial y} - 0 \right) - \hat{y} \left(\frac{\partial A_z}{\partial x} - 0 \right) + \hat{z}(0 - 0), \\ \vec{B} &= \hat{x}(2A_0y) - \hat{y}(2A_0x) = 2A_0(y\hat{x} - x\hat{y}).\end{aligned}$$

Note that in cylindrical coordinates, $x = r \cos \phi$ and $y = r \sin \phi$, so:

$$y\hat{x} - x\hat{y} = r \sin \phi \hat{x} - r \cos \phi \hat{y} = -r \hat{\phi},$$

where we used $\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$. Therefore,

$$\vec{B} = -2A_0r \hat{\phi}.$$

This points in the $\hat{\phi}$ direction (around the z axis), consistent with the right-hand rule for current in the $+\hat{z}$ direction. However, $A_0 < 0$. From problem 1, the magnetic field inside a wire carrying uniform current is

$$\vec{B} = \frac{\mu_0 I r}{2\pi r_0^2} \hat{\phi}.$$

Setting the two expressions equal gives

$$-2A_0r = \frac{\mu_0 I r}{2\pi r_0^2} \implies \boxed{A_0 = -\frac{\mu_0 I}{4\pi r_0^2}}.$$

□

Question 5. Purcell chapter 6, exercise 6.49 (Field at the center of a disk.) A disk with radius R and surface charge density σ spins with angular frequency ω . What is the magnetic field at the center?

Solution. Consider a thin ring of radius r and width dr on the disk, and assume the area is oriented such that $\vec{n}$ points in the $+\vec{z}$ direction. The charge on this ring is

$$dq = \sigma \cdot 2\pi r \, dr.$$

The ring completes one revolution in period $T = 2\pi/\omega$, so the current carried by this ring is

$$dI = \frac{dq}{T} = \frac{\sigma \cdot 2\pi r \, dr}{2\pi/\omega} = \sigma \omega r \, dr.$$

Using the Biot–Savart law, at the center of the ring, every element $d\vec{\ell}$ is perpendicular to $\hat{r}$, so $|d\vec{\ell} \times \hat{r}| = d\ell$. Integrating around the full loop:

$$dB = \frac{\mu_0}{4\pi} \frac{dI}{r^2} \oint d\ell = \frac{\mu_0}{4\pi} \frac{dI}{r^2} (2\pi r) = \frac{\mu_0}{2\pi} \frac{dI}{r} = \frac{\mu_0 \sigma \omega r \, dr}{2\pi} = \frac{\mu_0 \sigma \omega}{2} dr.$$

Integrating from 0 to R gives

$$B = \int_0^R \frac{\mu_0 \sigma \omega}{2} dr = \frac{\mu_0 \sigma \omega}{2} R.$$

The field points along the axis of rotation by the right-hand rule, so

$$\boxed{\vec{B} = \frac{\mu_0 \sigma \omega R}{2} \hat{z}.$$

□

Question 6. A classic infinite current sheet carrying a current/length λ (for example, in the $-\hat{y}$ direction) creates constant magnetic fields on either side of the sheet whose magnitudes are equal, but whose directions are 180° apart.

- (a) What direction and magnitude of a constant $\vec{B}$ field needs to be superimposed on this so that the B field still is constant in magnitude above and below the sheet, but now has a 90° change in direction when comparing above and below the current sheet?
- (b) How could you produce the extra constant B field required to make (a) work?

Solution. The infinite current sheet with current per length λ in the $-\hat{y}$ direction produces

$$\vec{B}_{\text{above}} = -\frac{\mu_0\lambda}{2}\hat{x}, \quad \vec{B}_{\text{below}} = \frac{\mu_0\lambda}{2}\hat{x}.$$

These are parallel, given $-\hat{x} \cdot \hat{x} = -1$, with equal magnitude. For part (a), superimpose a constant field

$$\vec{B}_{\text{ext}} = \frac{\mu_0\lambda}{2}\hat{y}.$$

Then,

$$\vec{B}_{\text{above}} = \frac{\mu_0\lambda}{2}(\hat{x} + \hat{y}), \quad \vec{B}_{\text{below}} = -\frac{\mu_0\lambda}{2}(\hat{x} - \hat{y}).$$

Both have the same magnitude

$$|\vec{B}| = \sqrt{\left(\frac{\mu_0\lambda}{2}\right)^2 + \left(\frac{\mu_0\lambda}{2}\right)^2} = \frac{\mu_0\lambda}{\sqrt{2}}.$$

We can check orthogonality by taking the dot product as such

$$(\hat{x} + \hat{y}) \cdot -(\hat{x} - \hat{y}) = (1, 1, 0) \cdot (-1, 1, 0) = 1(-1) + 1(1) + 0 = 0.$$

Since the dot product is zero, the fields above and below are 90° apart. For part (b), place two parallel infinite current sheets, one above and one below the original sheet. The upper sheet carries current in the $+\hat{x}$ direction and the lower sheet carries current in the $-\hat{x}$ direction, both with current per length $\lambda/2$. Between the two sheets, their fields superimpose to produce a uniform $\vec{B} = \frac{\mu_0\lambda}{2}\hat{y}$, which is exactly the constant field needed. $\square$

Question 7. Purcell chapter 7, exercise 7.44 (Magnetic energy in the galaxy.) A magnetic field exists in most of the interstellar space in our galaxy. There is evidence that its strength in most regions is between 10^{-6} and 10^{-5} gauss. Adopting 3×10^{-6} gauss as a typical value, find, to order of magnitude, the total energy stored in the magnetic field of the galaxy. For this purpose you may assume the galaxy is a disk roughly 10^{21} m in diameter and 10^{19} m thick. To see whether the magnetic energy amounts to much, on that scale, you might consider the fact that all the stars in the galaxy are radiating about 10^{37} joules/second. How many years of starlight is the magnetic energy worth?

Solution. First, convert the magnetic field to SI units so that

$$B = 3 \times 10^{-6} \text{ gauss} = 3 \times 10^{-10} \text{ T}.$$

The volume of the galaxy, modeled as a disk of diameter 10^{21} m (radius 5×10^{20} m) and thickness 10^{19} m, is

$$V = \pi r^2 h = \pi (5 \times 10^{20} \text{ m})^2 (10^{19} \text{ m}) \approx 8 \times 10^{60} \text{ m}^3.$$

The energy density of the magnetic field is

$$u = \frac{B^2}{2\mu_0} = \frac{(3 \times 10^{-10} \text{ T})^2}{2(4\pi \times 10^{-7} \text{ T} \cdot \text{m} \cdot \text{A}^{-1})} \approx 3.6 \times 10^{-14} \text{ J/m}^3.$$

Therefore, the total energy stored in the magnetic field is

$$U = u \cdot V \approx (3.6 \times 10^{-14} \text{ J/m}^3)(8 \times 10^{60} \text{ m}^3) \approx 3 \times 10^{47} \text{ J}.$$

To find how many years of starlight this is equivalent to, divide

$$t = \frac{U}{P} = \frac{3 \times 10^{47} \text{ J}}{10^{37} \text{ J/s}} = 3 \times 10^{10} \text{ s} \implies \frac{3 \times 10^{10} \text{ s}}{3.15 \times 10^7 \text{ s/yr}} \approx 10^3 \text{ years}.$$

□